SCIENTIFIC PORTFOLIO

AKONASSOU SEDJRO TCHÉDRAC SINGBO

Medical Biotechnology · Laboratory practice · Biomedical data

Profile

Biomedical laboratory professional in Cotonou, Benin. The portfolio brings together reported laboratory experience, an undergraduate thesis, and reproducible technical demonstrations on clearly labelled synthetic data. The technical projects are not clinical results and are not presented as hospital work.

Education and thesis

Medical Biotechnology programme, École Nationale Supérieure des Biosciences et Biotechnologies Appliquées (ENSBBA–UNSTIM), Benin. The supplied thesis cover identifies a professional-degree thesis and records a defence on 23 July 2024. The issued diploma itself was not supplied.

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Professional experience

Period	Experience reported in the portfolio material
Since Jul 2026	Biomedical laboratory activity · CHU de la Mère et de l’Enfant (CHU-MEL), Cotonou. The exact job title is not stated here.
Jun–Sep 2025	Medical laboratory internship · CHUD Ouémé-Plateau, Porto-Novo.
Mar–Jun 2024	Thesis work · CHUD Ouémé-Plateau; the thesis documents the study and its laboratory analyses.
Jul–Aug 2023	Academic internship · Centre de Santé d’Adjarra.

Experience dates and descriptions above are carried over from the supplied portfolio material; employment certificates were not included for independent verification. The official CHU-MEL job title remains to be confirmed.

TECHNICAL PORTFOLIO

Reproducible demonstrations

Projects 01–04 use synthetic datasets. Each project includes source files and scripts; its outputs illustrate computation and data checks, not clinical or surveillance findings.

Project	Scope and evidence	Status
01 · Antimicrobial resistance	100 synthetic observations; 10 organism–antibiotic pairs, 10 observations each. S/I/R categories remain separate; R proportions use S+I+R as denominator.	Synthetic demonstration
02 · Sequence quality control	Three synthetic FASTA sequences (191 bases total; 8 ambiguous bases) and three short synthetic FASTQ reads. Reports sequence/read quality indicators and exploratory ORFs.	Synthetic demonstration
03 · SQL and ETL	191 source rows; 191 curated rows, 6 groups and no anomalies in the main fixture. A separate five-row test fixture checks missing, invalid and duplicate records.	Synthetic demonstration
04 · Statistical analysis	300 synthetic records in six groups (n=50 each). Reports resistance proportions and 95% Wilson intervals; denominator is valid S+I+R results.	Synthetic demonstration

Skills demonstrated by the repository

Standard-library Python for CSV/JSON processing and validation; FASTA/FASTQ parsing and basic quality metrics; SQL and SQLite relational constraints/aggregations; descriptive proportions and Wilson intervals; SHA-256 file fingerprinting; SVG chart generation; and static HTML/CSS project pages.

Tools not demonstrated in delivered project code

No operational Power BI dashboard or .pbix file is included. The project scripts do not demonstrate pandas, Biopython or Matplotlib. Hospital practice and the technical portfolio are kept distinct.

Open project pages

01 · project_01_antimicrobial_resistance 02 · project_02_bacterial_sequence
03 · project_03_sql_advanced 04 · project_04_advanced_biomedical_analysis
05 · project_05_memoire_soutenance_prostate

THESIS SUMMARY

Evaluation of inflammation in prostatic dysfunction

Thesis title translated from the supplied French document · CHUD-OP · defence: 23 July 2024

Study description reported in the thesis

The document describes 81 men aged 40 years or older: 61 retrospective records (1 January 2023–20 March 2024) and 20 prospective participants recruited from dialysis care (25 March–22 June 2024). The assessed markers were total PSA, C-reactive protein (CRP) and complete blood count (CBC/NFS), including leucocytes, neutrophils and lymphocytes. No cytokine assay is reported; cytokines appear only in the literature discussion.

The thesis reports means ± standard error of the mean, Mann–Whitney comparisons and a 5% significance threshold. The underlying individual-level dataset and source tables were not supplied, so the analysis cannot be independently reproduced.

Aggregate results as stated in the thesis

Marker	Normal PSA (n=69)	Elevated PSA (n=12)	Reported comparison
Leucocytes (G/L)	5.39 ± 0.46	5.61 ± 0.92	Not significant
Neutrophils (G/L)	2.73 ± 0.25	2.94 ± 0.67	Not significant
Lymphocytes (G/L)	1.87 ± 0.09	2.12 ± 0.17	$p < 0.05$
CRP (mg/L)	4.61 ± 0.63	4.48 ± 1.50	Not significant

Interpretation and limits

These are values and significance labels reported by the thesis, not recalculated results. Exact p-values, confidence intervals, correlation outputs and individual distributions are unavailable. The reported association between PSA group and lymphocyte count does not establish an inflammatory mechanism or causality. No academic mark or honours classification is included because the supplied source does not document one.